

Software Requirements Specification

for

Peer to Peer Skill Exchange

Version 1.0 approved

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Revision History

Name Date Reason For Changes Version			

1. Introduction

1.1 Purpose

Peer to Peer Skill Exchange Software is a platform for students to interact and exchange their skills. A user can be, both a student and teacher and can list their skills to get matched with their compatible peers. A user can earn credits by teaching and spend them for learning. Skill matching can be done based on skills, ratings and consistency.

1.2 Document Conventions

The SRS document follows IEEE 830 standard.

1.3 Intended Audience and Reading Suggestions

This document is intended for the following stakeholders:

- **Developers:** *To understand the functional and technical requirements for implementation.*
- **Project Managers:** *To monitor project scope, planning, and progress.*
- **System Testers/QA Engineers:** *To create and execute test cases based on the specified requirements.*
- **UI/UX Designers:** *To design an intuitive and user-friendly interface.*
- **End Users (Students):** *To understand the features and capabilities of the platform.*
- **Documentation Team:** *To prepare user manuals and technical documentation.*

It is in the order of introduction, description, specific requirements and appendices.

1.4 Product Scope

The software aims to allow the user to learn and interact with people with similar interests. Credit wallets can be used to reward the user and encourage them to participate and engage in upskilling sessions of their choice.

2. Overall Description

2.1 Product Perspective

The idea of Peer-to-Peer Skill Exchange Platform was conceived to solve the issues and problems faced by students to find economically accessible and good quality tutoring from trustworthy sources. The “Credit-Wallet” feature makes skill learning accessible to students through peer-to-peer exchange rather than paid courses.

This is a self-contained product that acts as a bridge between a learner(a student) and a tutor(also can be a student).The users can list down the skills they are interested in and the skills that they have.It allows the users to schedule or cancel the sessions based on the availability and skill matching.All transactions on this platform are done in terms of “Credits”.Credits can be earned by teaching . A user can learn and pay for the session using the credits earned.This helps in creating a sustainable learning ecosystem.A single registered user can switch between the roles of a meteor and a learner.

2.2 Product Functions

Some of the major functions of this product are:

1. User Authentication & Profile

- **Auth:** Register/login securely via email/phone, password reset, and secure logout.
- **Profile:** Manage personal details, upload profile pictures, define teachable/learning skills, and set experience levels (Beginner, Intermediate, Advanced).

2. Skill Management & Discovery

- **Listings:** Add, modify, delete, and browse available skills to teach.
- **Search & Match:** Keyword search with filters (category, level, ratings, availability) to recommend mentors.

3. Sessions & Wallet

- **Scheduling:** Send, accept/reject, cancel, or reschedule session requests without the problem of double booking.
- **Wallet:** Virtual credit system (credits awarded to mentors and deducted from learners post-session) with transaction history.

4. Feedback & Notifications

- **Ratings:** Post-session star ratings, written reviews, and automated mentor average calculations.
- **Alerts:** Notifications for requests, status updates, upcoming sessions, cancellations, and rescheduling.

5. Administration

- **Admin Tools:** Manage user accounts, moderate listings, resolve disputes, handle reports, and generate system analytics.

2.3 User Classes and Characteristics

The User classes are:

1. *Guest(Unregistered User):Can Browse the Platform,search available skills(limited view),register for an account.*
2. *Registered User(Learner/Mentor):Can manage profile,list skills,search for skills,request sessions,earn and spend credits,rate other users.*
3. *Learner (role of a registered user):Can search mentors,book sessions,spend credits,attend sessions,provide ratings and reviews.*
4. *Mentor (role of a registered user):Can create skill listings,accept or reject requests,conduct sessions,earn credits,receive ratings and reviews.*
5. *Administrator: Can verify users(if applicable),manage accounts,remove inappropriate content,resolve disputes,monitor transactions,generate reports.*

2.4 Operating Environment

The Peer-to-Peer Skill Exchange is a web based application designed to operate in a standard client-server environment.The system should be accessible from desktop,laptops,tablets smartphones through web browsers.

1.Hardware platform

Client Side

- *Desktop or Laptop with a minimum dual-core processor and 4 GB RAM.*
- *Smartphone or Tablet with Android or iOS operating systems.*
- *Stable internet connection (Wi-Fi or mobile data).*

Server Side

- *Cloud or local server with at least:*
 - *Quad-core processor*
 - *8 GB RAM*
 - *100 GB storage (expandable as needed)*
- *Reliable internet connectivity for hosting the application and database.*

Operating Systems

- *Windows 10/11*
- *Linux*

Web Browsers

The application should function correctly on the latest stable versions of:

- *Google Chrome*
- *Microsoft Edge*

- *Mozilla Firefox*

2.5 Design and Implementation Constraints

- The System will be built as a web application that can run on modern web browsers.
- The project must be completed within the semester timeline using Agile sprints.
- The application should work on Windows and Linux
- The Project will use HTML,CSS,JavaScript,React.js,Node.js/Express.js and MongoDB
- The system
- should use HTTPS to ensure secure communication and protect the user data.
- User passwords must be stored in encrypted (hashed) form.
- The platform should support multiple users accessing it simultaneously without affecting the performance.
- The user interface should be simple,responsive and easy to use on both desktop and mobile devices.
- The code should follow standard programming practices and be maintained using Git and GitHub for version control.

3. External Interface Requirements

3.1 User Interfaces

General UI Requirements

- *Responsive web interface that works on desktops.*
- *Clean dashboard with easy navigation*
- *Consistent colour theme.*
- *Role- based dashboard(Student,Teacher and Admin)*
- *Support both light mode and dark mode.*

Authentication:

- *Login Page*
- *Registration Page*
- *Forgot Page Password*
- *OTP verification screen*

User Dashboard:

- *User profile*
- *Credit balance*
- *Upcoming sessions*
- *Recommended mentors*
- *Notifications*

Profile Management

Allows user to:

- *Upload profile picture*
- *Edit their bio*
- *Add skills to teach*
- *Add skills to learn*
- *Set availability calender*
- *Mention experience level*

Search Interface

- *Search bar*
- *Category filters*
- *Rating filters*
- *Availability filters*
- *Experience filters*
- *Sort by relevance*

Session Interface

- *Session request popup*
- *Calendar view*
- *Accept/Reject buttons*
- *Cancel/Reschedule options*
- *Session history*

Credit Wallet

- *Current balance*
- *Credit transaction history*
- *Credits earned*
- *Credits spent*

Rating Interface

- *Star rating(1-5)*
- *Written review*
- *Report inappropriate behaviour*

Notification Center

- *Session reminders*
- *Accepted requests*
- *Credit updates*
- *Admin announcement*

Admin Panel

- *User management*
- *Skill moderation*
- *Report management*
- *Session statistics*
- *Credit transaction monitoring*

3.2 Hardware Interfaces

This is mainly a web application so hardware requirements are minimal.

- *Desktop/Laptop*
- *Internet connection*
- *Webcam and microphone*
- *Speak or Headphones*

3.3 Software Interfaces

Operating Systems

- *Windows 10/11*
- *Linux*

Web Browsers

- *Google Chrome*
- *Microsoft Edge*
- *Mozilla Firefox*

Database

- *MongoDB*

Backend

- *Node.js*
- *Express.js*

Frontend

- *React.js*

Authentication

- *JWT Authentication*

APIs

- *Email API*
- *Calendar API*
- *Video Meeting API*

3.4 Communications Interfaces

Communication Protocols

- *HTTP/HTTPS*
- *REST APIs*
- *JSON data format*

Network Requirements

- *Stable internet connection*
- *HTTPS for secure communication*

Notification

- *Email notification*
- *In app notifications*
- *Push notifications*

Security

- *TLS encryption*
- *Password hashing using bcrypt*
- *JWT tokens for authenticated requests*
- *HTTPS communication for sensitive data*

4. System Features

4.1 User Registration and Login

4.1.1 Description and Priority

Description: This feature allows new users to create an account on the platform using their email address or phone number. Registered users can log in to access the platform, reset their password if forgotten, and log out when they finish using the system. User authentication ensures that only authorized users can access personal information, sessions, and credit wallets.

Priority: High

4.1.2 Stimulus/Response Sequences

- The user enters registration details.
- The system creates the account.
- The user logs in.
- The system opens the dashboard.

4.1.3 Functional Requirements

REQ-1: The system shall allow users to register.

REQ-2: The system shall allow users to log in.

REQ-3: The system shall allow users to log out.

4.2 User Profile

4.2.1 Description and Priority

Description: This feature allows new users to create an account on the Peer-to-Peer Skill Exchange platform using their email address or phone number. Registered users can securely log in to access the platform, reset their password if forgotten, and log out when they finish using the system. User authentication ensures that only authorized users can access personal information, sessions, and credit wallets.

Priority: High

4.2.2 Stimulus/Response Sequences

- The User opens their profile.
- The User edits their profile.
- The system saves the changes.

4.2.3 Functional Requirements

REQ-4: *The system shall allow users to edit their profile.*

REQ-5: *The system shall allow users to add skills to teach and learn.*

REQ-6: *The system shall allow users to upload a profile picture.*

4.3 Skill Search and Matching

4.3.1 Description and Priority

Description: *This feature helps users search for skills and find suitable mentors. Users can search using keywords and apply filters such as skill category, ratings, availability, and experience level. The system recommends mentors whose skills and availability match the learner's requirements.*

Priority: *High*

4.3.2 Stimulus/Response Sequences

- *The User searches for a skill.*
- *The System displays matching mentors.*
- *The User selects a mentor.*

4.3.3 Functional Requirements

REQ-9: *The system shall allow users to search for skills.*

REQ-10: *The system shall allow users to filter search results.*

REQ-11: *The system shall display matching mentors.*

4.4 Session Management

4.4.1 Description and Priority

Description: *This feature allows learners to request learning sessions and mentors to accept or reject those requests. Users can also reschedule or cancel sessions.*

Priority: *High*

4.4.2 Stimulus/Response Sequences

- *The learner requests a session.*
- *The mentor accepts or rejects the request.*
- *The system updates the session schedule.*

- *Notifications are sent to both users.*

4.4.3 Functional Requirements

- **REQ-13:** *The system shall allow learners to request sessions.*
- **REQ-14:** *The system shall allow mentors to accept or reject requests.*
- **REQ-15:** *The system shall allow users to cancel sessions.*
- **REQ-16:** *The system shall allow users to reschedule sessions.*
- **REQ-17:** *The system shall prevent double booking of mentors.*

4.5 Credit Wallet Management

4.5.1 Description and Priority

Description: *This feature maintains a virtual credit wallet for every user. Credits are earned by teaching sessions and spent when booking sessions.*

Priority: *High*

4.5.2 Stimulus/Response Sequences

- *A session is completed.*
- *The system transfers credits.*
- *Wallet balances are updated.*
- *The transaction is recorded.*

4.5.3 Functional Requirements

- **REQ-18:** *The system shall maintain a virtual credit wallet.*
- **REQ-19:** *The system shall add credits after successful teaching sessions.*
- **REQ-20:** *The system shall deduct credits when a learner books a session.*
- **REQ-21:** *The system shall display wallet balance.*
- **REQ-22:** *The system shall maintain transaction history.*

4.6 Ratings and Reviews

4.6.1 Description and Priority

Description: *After completing a session, learners can rate mentors and submit reviews to help other users choose suitable mentors.*

Priority: *Medium*

4.6.2 Stimulus/Response Sequences

- *The learner completes a session.*
- *The learner submits a rating.*
- *The system stores the review.*
- *The mentor's average rating is updated.*

4.6.3 Functional Requirements

- **REQ-23:** *The system shall allow learners to rate mentors.*
- **REQ-24:** *The system shall allow learners to submit reviews.*
- **REQ-25:** *The system shall calculate average ratings.*
- **REQ-26:** *The system shall display mentor ratings.*

4.7 Notifications

4.7.1 Description and Priority

Description: *The system sends notifications regarding session requests, session reminders, cancellations, wallet updates, and important announcements.*

Priority: *Medium*

4.7.2 Stimulus/Response Sequences

- *A user performs an action.*
- *The system generates a notification.*
- *The recipient receives the notification.*

4.7.3 Functional Requirements

- **REQ-27:** *The system shall notify users of new session requests.*
- **REQ-28:** *The system shall notify users when requests are accepted or rejected.*
- **REQ-29:** *The system shall send reminders before scheduled sessions.*
- **REQ-30:** *The system shall notify users about wallet transactions.*

4.8 Administration

4.8.1 Description and Priority

Description: *This feature enables administrators to monitor and manage the overall platform, including user accounts, reported content, skill listings, and system reports.*

Priority: *High*

4.8.2 Stimulus/Response Sequences

- *The administrator logs in.*
- *The administrator views system information.*
- *The administrator performs management actions.*
- *The system updates the database.*

4.8.3 Functional Requirements

- **REQ-31:** The system shall allow administrators to manage user accounts.
- **REQ-32:** The system shall allow administrators to remove inappropriate skill listings.
- **REQ-33:** The system shall allow administrators to resolve reported issues.
- **REQ-34:** The system shall generate reports on users, sessions, and credit transactions.
- **REQ-35:** The system shall monitor platform activities.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- *The website should load the dashboard and search results within 2-3 seconds.*
- *When a user searches for a skill or mentor, results should show up quickly, without much delay.*
- *Booking, accepting or cancelling a session should update on both users' screens almost immediately.*
- *Credit balance should update right after a session is completed, so users can see it without refreshing multiple times.*
- *The system should be able to handle a reasonable number of users at the same time without slowing down.*

5.2 Safety Requirements

- *The system should not allow two learners to book the same mentor for the same time slot.*
- *Credits should be deducted from the learner and added to the mentor only after the session is properly marked as completed, so there is no mismatch in the wallet balance.*
- *If something goes wrong while a transaction is happening (like the app crashing), the credit balance should not get corrupted or updated incorrectly.*
- *All credit transactions should be recorded so that if there is any issue later, it can be checked and resolved by the admin.*

5.3 Security Requirements

- *Passwords should never be stored as plain text; they should be stored using hashing (bcrypt).*
- *Login sessions should be handled using JWT tokens so that users stay authenticated securely.*
- *OTP verification should be used at the time of registration/login to confirm that the user is genuine.*
- *All sensitive data (like login details and credit transactions) should be sent using HTTPS so that it cannot be easily intercepted.*
- *Users should only be able to access features meant for their role — a normal user should not be able to access the Admin Panel.*
- *A user's personal details (like phone number or email) should not be visible to other users unless required.*

5.4 Software Quality Attributes

- **Usability:** *The website should be easy to use even for a first-time user — a student should be able to register, list a skill and book a session without much confusion.*

- **Reliability:** *The system should work correctly most of the time, especially for important actions like booking sessions and credit transactions.*
- **Maintainability:** *The code should be organised in a proper way (separate parts for login, sessions, credits, ratings, etc.) so that it is easy to fix bugs or add new features later.*
- **Portability:** *Since it is a web application, it should work properly on common browsers like Chrome, Edge and Firefox.*
- **Scalability:** *The system should be built in a way that it can support more users in the future without needing to redesign everything from scratch.*

5.5 Business Rules

- *Only registered users can book sessions, earn/spend credits or rate other users. Guests can only view limited content.*
- *A single user can act as both a learner and a mentor, and can switch between the two roles.*
- *Credits can only be earned by teaching a session and spent by booking a session — credits are not bought using real money.*
- *A learner must have enough credits in their wallet to book a session; otherwise the booking should not be allowed.*
- *A learner can rate and review a mentor only after the session is completed, and only once for each session.*
- *Only the admin can remove inappropriate skill listings, resolve disputes or take action on reported users.*

6. Other Requirements

Appendix A: Glossary

<Define all the terms necessary to properly interpret the SRS, including acronyms and abbreviations. You may wish to build a separate glossary that spans multiple projects or the entire organization, and just include terms specific to a single project in each SRS.>

Appendix B: Analysis Models

<Optionally, include any pertinent analysis models, such as data flow diagrams, class diagrams, state-transition diagrams, or entity-relationship diagrams.>

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Appendix C: To Be Determined List

<Collect a numbered list of the TBD (to be determined) references that remain in the SRS so they can be tracked to closure.>